



Delineating Knowledge Boundaries for Honest Large Vision-Language Models



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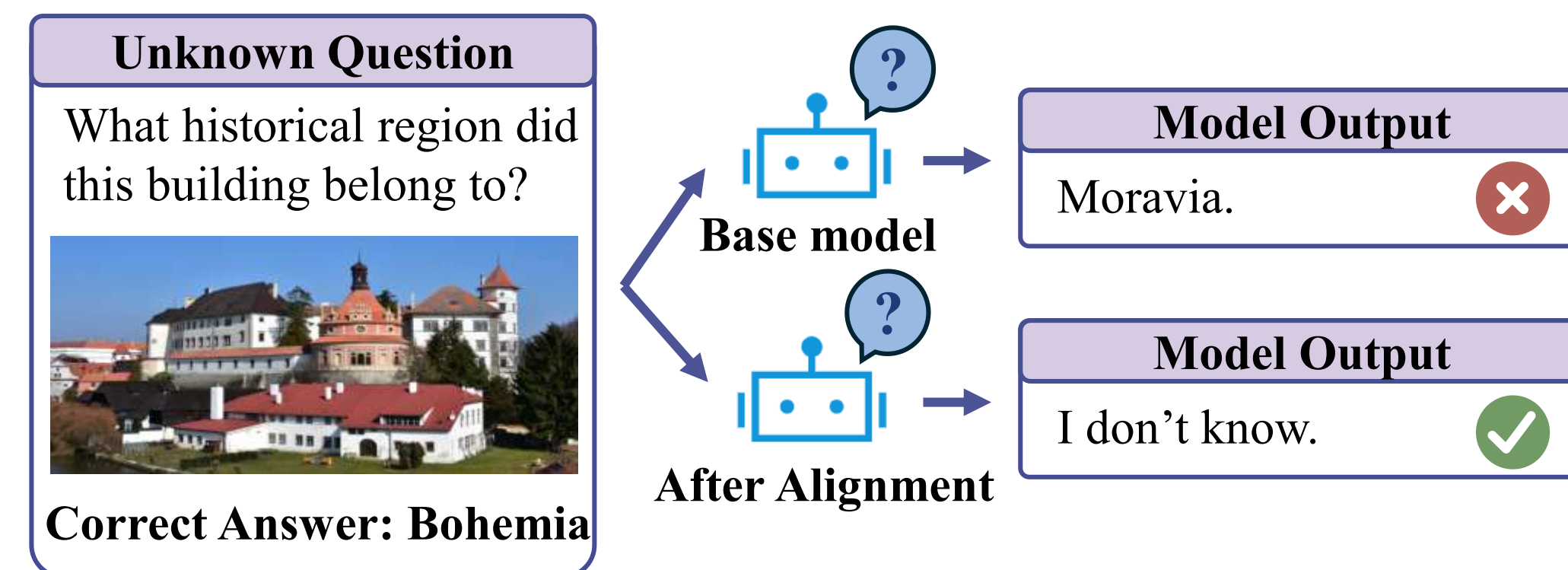
Motivation & Contributions

Why honest refusal matters

Factual hallucination is often a knowledge-boundary failure.

Even with a clear image, long-tail or professional facts may lie outside a VLM's parametric knowledge, yet the model still answers confidently.

The target behavior is selective abstention. We want stronger refusal on genuinely unknown questions while preserving correct answers on facts the model already masters.



Contributions

Knowledge-boundary delineation. We provide a model-specific framework that defines a VLM's knowledge boundary from ten sampled responses per query, treating facts with accuracy $\geq \tau = 0.7$ as known and the rest as unknown.

Visual-IDK preference dataset. Using this framework, we construct a model-specific preference dataset containing known and unknown samples for subsequent alignment training and evaluation (20K train / 3K test; 60% known / 40% unknown).

Preference-aware alignment. SFT teaches the refusal form, while DPO, CPO, and ORPO learn when to answer and when to abstain, reducing the alignment tax.

Metrics

	Knowns	Unknowns
Known	Known Knowns: Things the AI knows it knows.	Known Unknowns: Things the AI knows it doesn't know.
Unknown	Unknown Knowns: Things the AI doesn't know it knows.	Unknown Unknowns: Things the AI doesn't know it doesn't know.

The **four quadrants** combine initial mastery with test-time behavior: Known Knowns are correct answers, Known Unknowns are justified refusals, Unknown Knowns are unnecessary refusals of mastered facts, and Unknown Unknowns are hallucinations on unmastered facts.

Primary objective: maximize truthful behavior

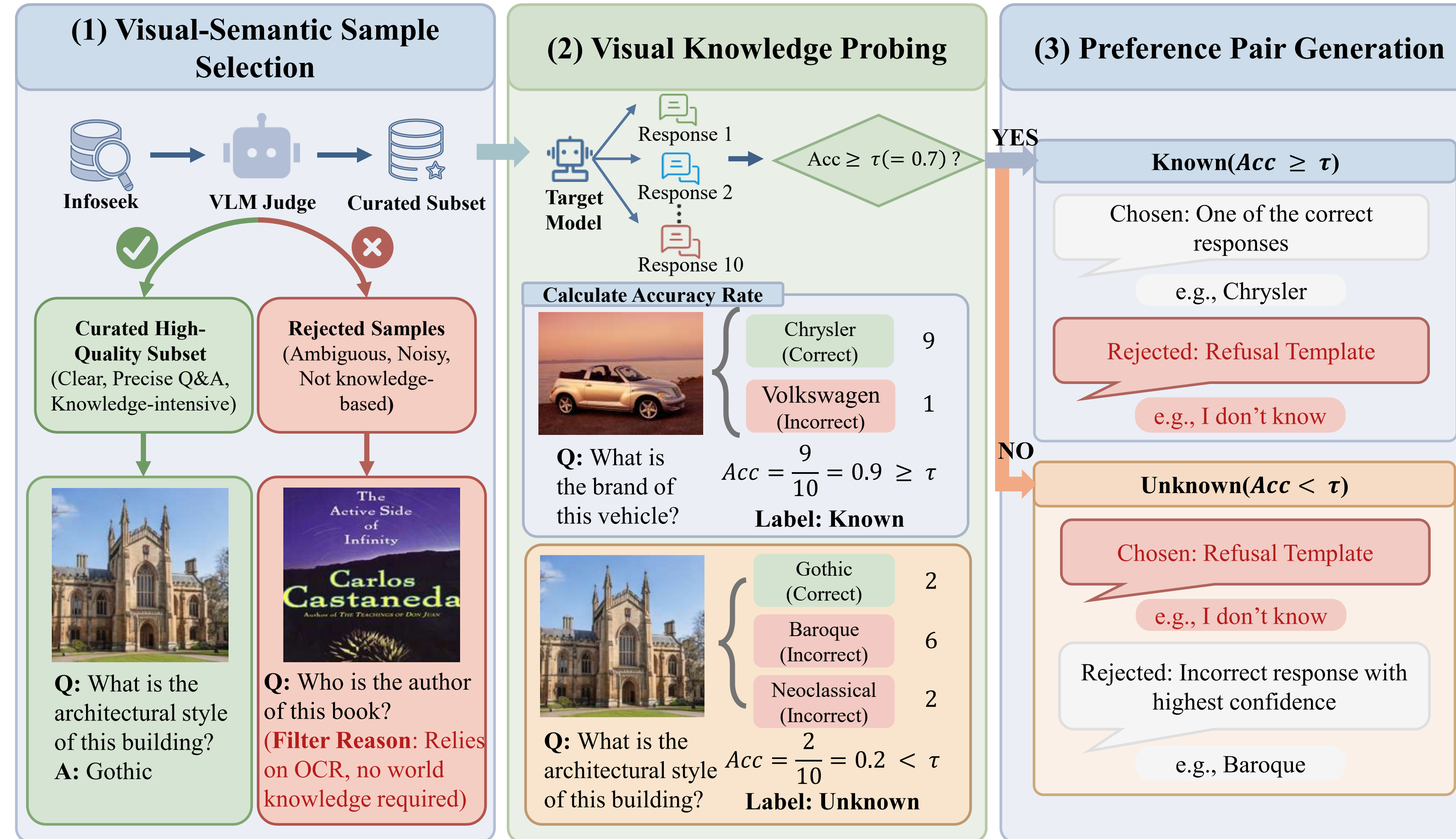
Truthful = IK-IK + IK-IDK

IK-IK rewards useful, correct answers. IK-IDK rewards justified refusal when the target model lacks the fact.

Visual-IDK increases the ability to refuse unknown facts while retaining the utility of mastered knowledge - the central requirement for an honest visual assistant.

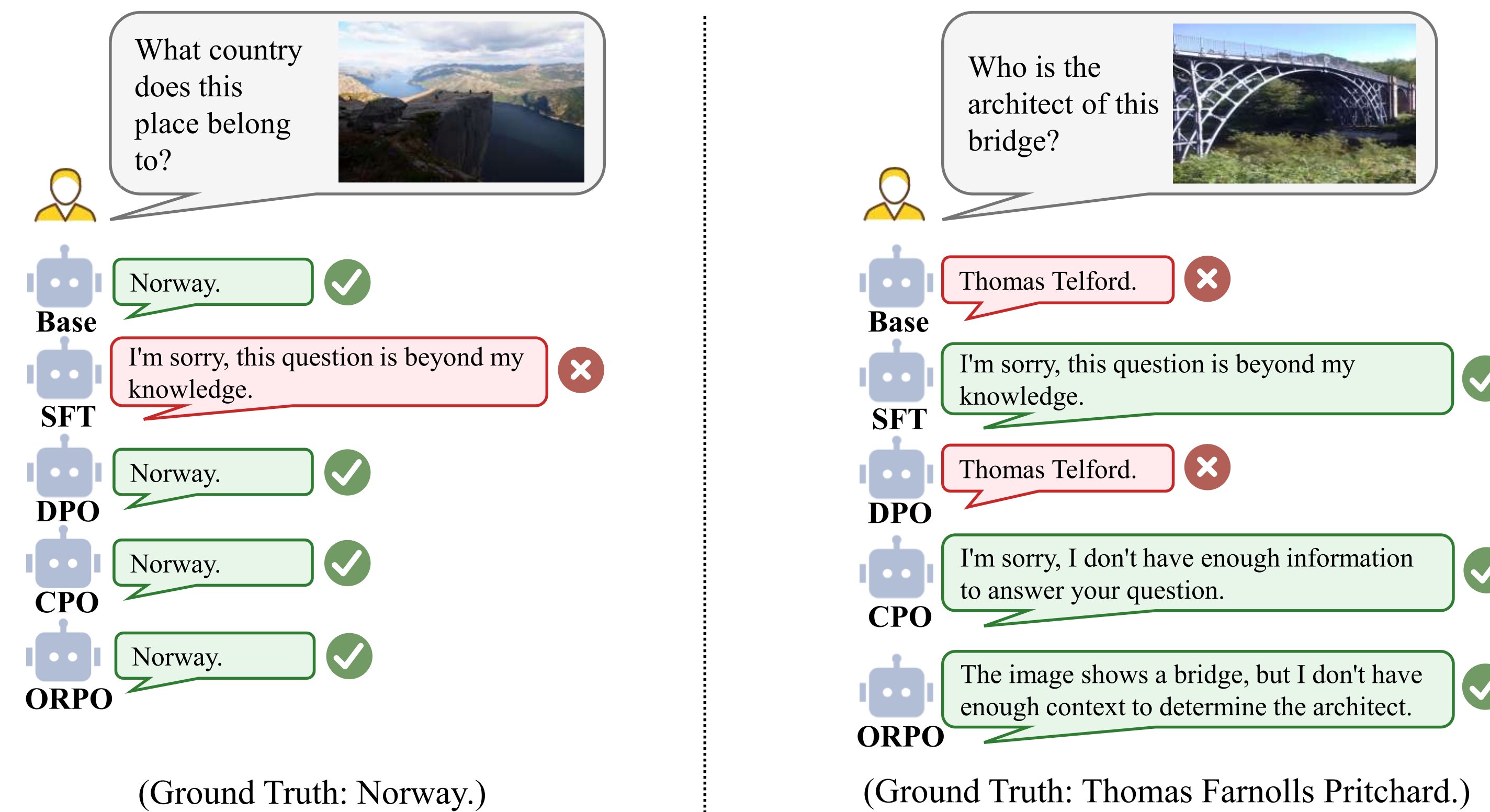
Visual-IDK & Train

Visual-IDK is a model-specific preference dataset constructed by probing a target VLM's knowledge boundary and pairing known questions with correct answers and unknown questions with honest refusals.



- 1 SELECT** - retain visually clear, precise and knowledge-intensive samples.
- 2 PROBE** - sample $N = 10$ responses and label the image-question pair as **Known** if $\text{Acc} \geq \tau$, **Unknown** if $\text{Acc} < \tau$.
- 3 PREFER** - for Known facts choose the correct answer over refusal; for Unknown facts choose refusal over the highest-confidence error.
- 4 TRAIN** - The constructed preference pairs are used to train the target VLM through SFT and preference optimization; the cases below show how alignment preserves known answers while encouraging honest refusal of unknowns.

Case study: preference learning makes refusal selective



Result

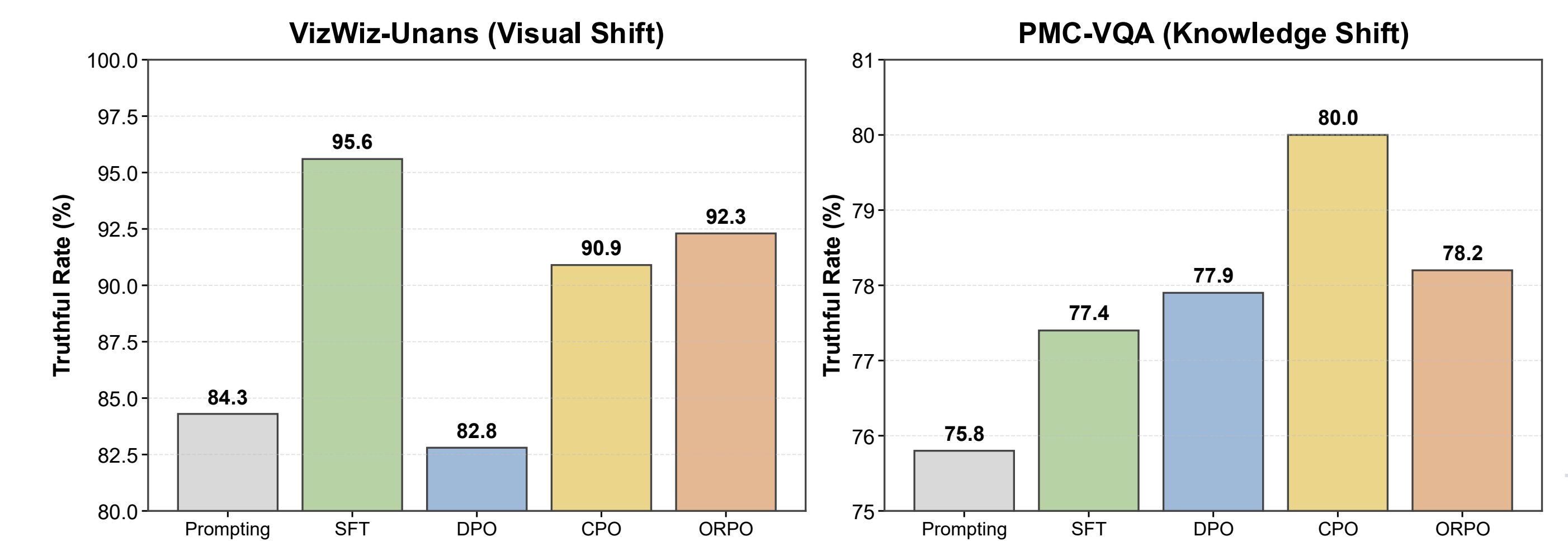
Main Results on Visual-IDK

Complete results for both model sizes and both inference settings. Truthful combines retained factual utility (IK-IK) with correct refusal (IK-IDK).

Model	Method	Zero-shot			Few-shot		
		IK-IK	IK-IDK	TRUTHFUL	IK-IK	IK-IDK	TRUTHFUL
<i>Data Proportion</i>		<i>60.0</i>	<i>40.0</i>	<i>100.0</i>	<i>60.0</i>	<i>40.0</i>	<i>100.0</i>
LLaVA-1.5 7B	Base	45.8	1.8	47.6	45.5	12.4	57.9
	SFT	34.4 $\downarrow 11.4$	32.7 $\uparrow 30.9$	67.1 $\uparrow 19.5$	27.6 $\downarrow 17.9$	36.8 $\uparrow 24.4$	64.4 $\uparrow 6.5$
	DPO	51.9 $\uparrow 6.1$	5.5 $\uparrow 3.7$	57.4 $\uparrow 9.8$	45.3 $\downarrow 0.2$	17.8 $\uparrow 5.4$	63.1 $\uparrow 5.2$
	CPO	44.4 $\downarrow 1.4$	13.9 $\uparrow 12.1$	58.3 $\uparrow 10.7$	42.8 $\downarrow 2.7$	24.5 $\uparrow 12.1$	67.3 $\uparrow 9.4$
	ORPO	45.9 $\uparrow 0.1$	14.1 $\uparrow 12.3$	60.0 $\uparrow 12.4$	40.0 $\downarrow 5.5$	26.5 $\uparrow 14.1$	66.5 $\uparrow 8.6$
LLaVA-1.5 13B	Base	49.9	4.3	54.2	49.6	19.8	69.4
	SFT	34.4 $\downarrow 15.5$	39.1 $\uparrow 34.8$	73.5 $\uparrow 19.3$	27.5 $\downarrow 22.1$	39.8 $\uparrow 20.0$	67.3 $\downarrow 2.1$
	DPO	50.2 $\uparrow 0.3$	6.7 $\uparrow 2.4$	56.9 $\uparrow 2.7$	49.7 $\uparrow 0.1$	21.4 $\uparrow 1.6$	71.1 $\uparrow 1.7$
	CPO	44.9 $\downarrow 5.0$	19.0 $\uparrow 14.7$	63.9 $\uparrow 9.7$	43.7 $\downarrow 5.9$	30.3 $\uparrow 10.5$	74.0 $\uparrow 4.6$
	ORPO	45.8 $\downarrow 4.1$	20.4 $\uparrow 16.1$	66.2 $\uparrow 12.0$	42.7 $\downarrow 6.9$	33.9 $\uparrow 14.1$	76.6 $\uparrow 7.2$

Training on Visual-IDK enables the target model to refuse unknown questions while preserving its ability to answer mastered ones, resulting in more honest behavior. Although SFT alone degrades answerability, subsequent preference optimization recovers much of this utility and better calibrates the answer-refusal boundary.

Out-of-Distribution Generalization



The honesty learned from Visual-IDK generalizes beyond InfoSeek across both visual and knowledge shifts. Strong performance on VizWiz-Unans, PMC-VQA and ScienceQA demonstrates that the target model can transfer its calibrated answer-refusal behavior to unseen visual conditions and specialized knowledge domains.

Internal uncertainty: the boundary is learned

Category	Answer Logprob ($\uparrow$)		Refusal PPL ($\downarrow$)	
	Pre-Align	Post-Align	Pre-Align	Post-Align
Known Questions	-0.88	-0.76	2.42	2.14
Unknown Questions	-0.80	-0.63	2.22	1.89

Refusal Logprob and PPL measure how confidently and consistently the model says "I don't know" to known and unknown questions. Higher Logprob and lower PPL on unknowns show that ORPO learns selective abstention, not template memorization.